Generative AI (GenAI) in Engineering Education:

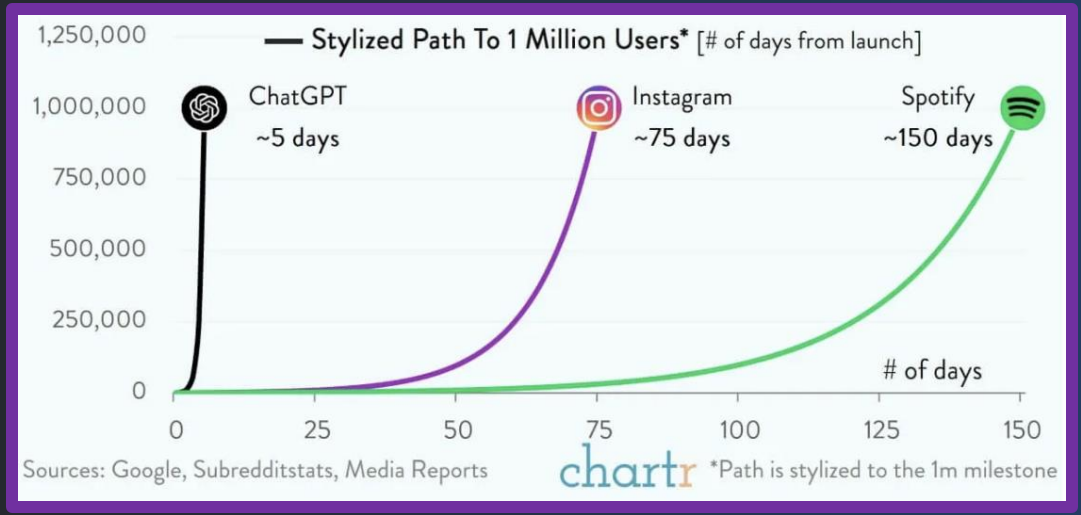
Insights from GPT-Classroom Implementation and Multi-Agent Learning Experiments

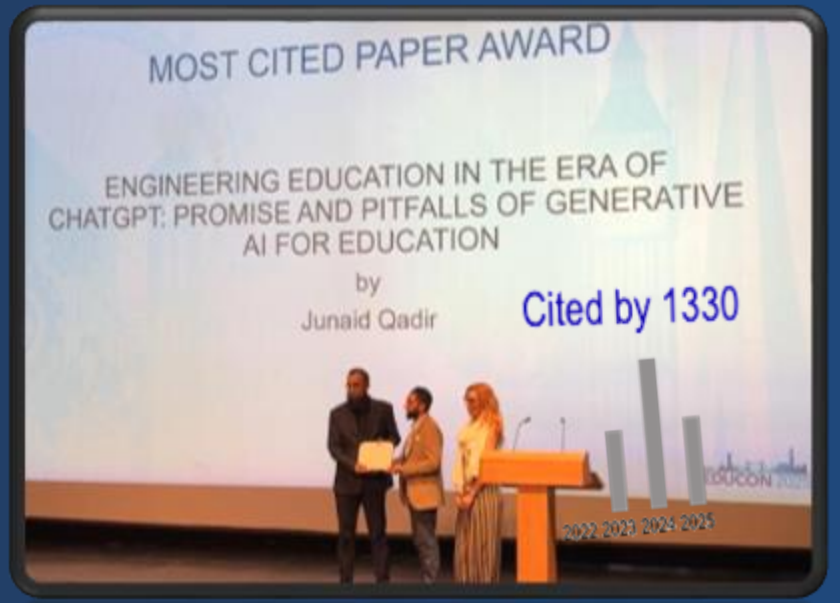
Prof. Junaid Qadir

College of Engineering, Qatar University, Doha, Qatar

1

The Exponential GenAI Educational Wave





Learning as an Infinite Game

First Law 

When the AI is readily accessible, your work should be no worse than that produced by the AI alone.
(Embrace change and leverage AI)

Second Law 

(Corollary of Law 1)
When AI is there, and you don't want to be made irrelevant, maintain your competitive advantage
(Become **AI Proof**)

Third Law 

After becoming users of AI, don't let your inherent unaided capabilities drop
(**Don't outsource your thinking muscle allowing it to atrophy**; i.e., do no harm

 PARADOX: The real challenge is doing the First and Third Laws simultaneously!

Paradoxes of Education

School–Work Contradiction (Warschauer et al.*)

GenAI is needed for future careers but discouraged in school, leaving students unprepared

EdTech Matthew Effect (Reich*)

GenAI widens inequality by benefiting resource-rich learners and advantaged learners more

Critical Thinking Contradiction (Hamilton, Wiliam, Hattie*)

GenAI requires critical thinking to use well; yet it may erode the very skills it depends on.

Productivity Contradiction & Irony of Automation (Simkute et al.*)

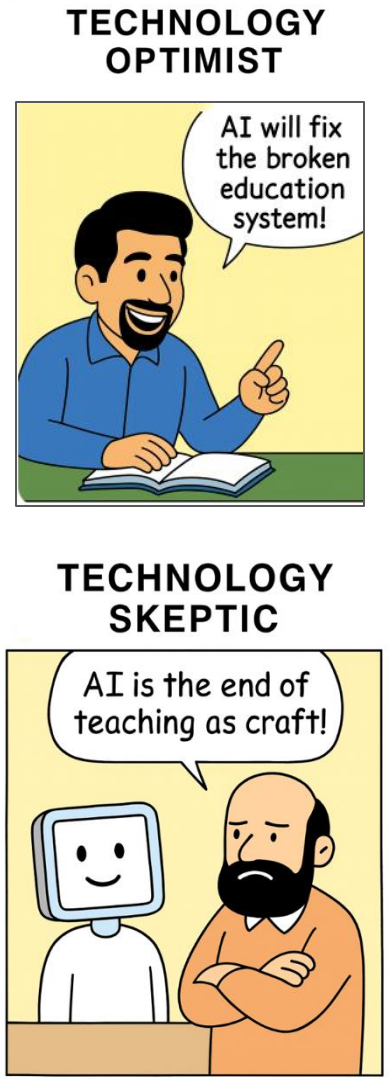
Automation may paradoxically demand more human oversight, increasing cognitive load.

** Full references available on request*

The Either-Or Approach

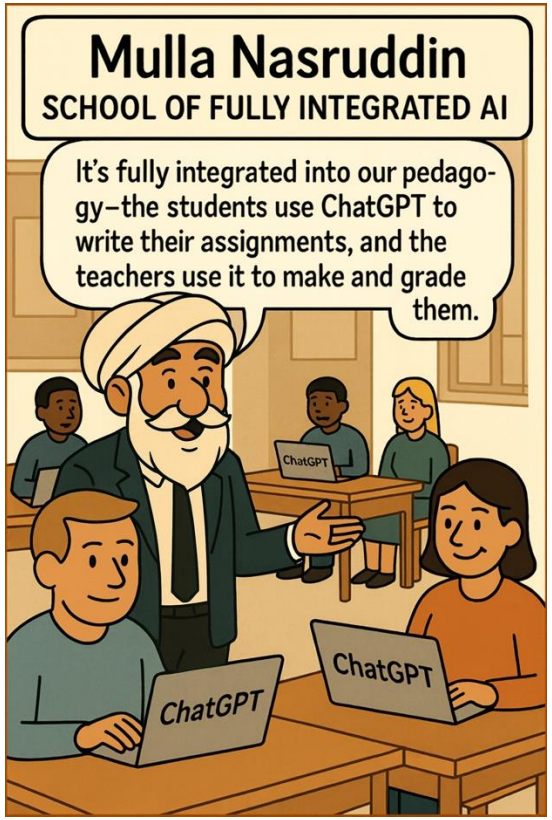
These protagonists are not entirely wrong—but their ***either-or view*** *is incomplete.*

We need a ***dynamic balance & synthesis***

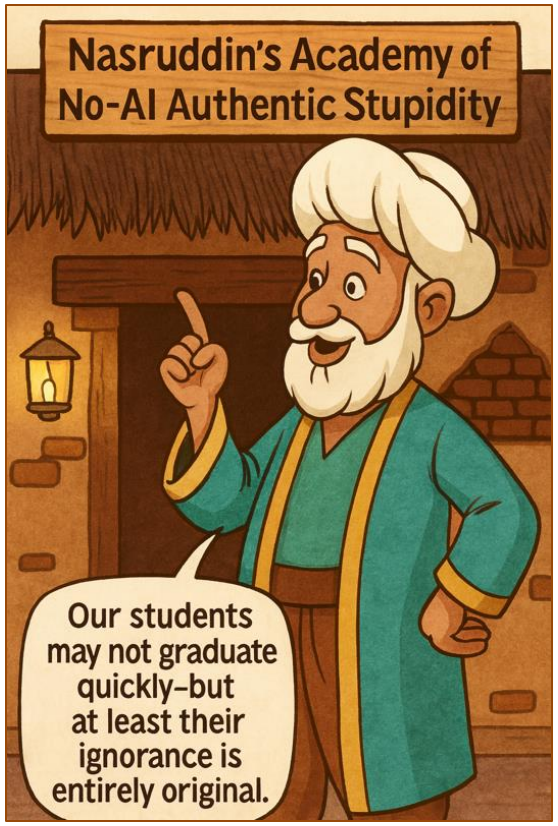


Two caricatures of bad implementations

FULL INTEGRATION



FULL SUBTRACTION

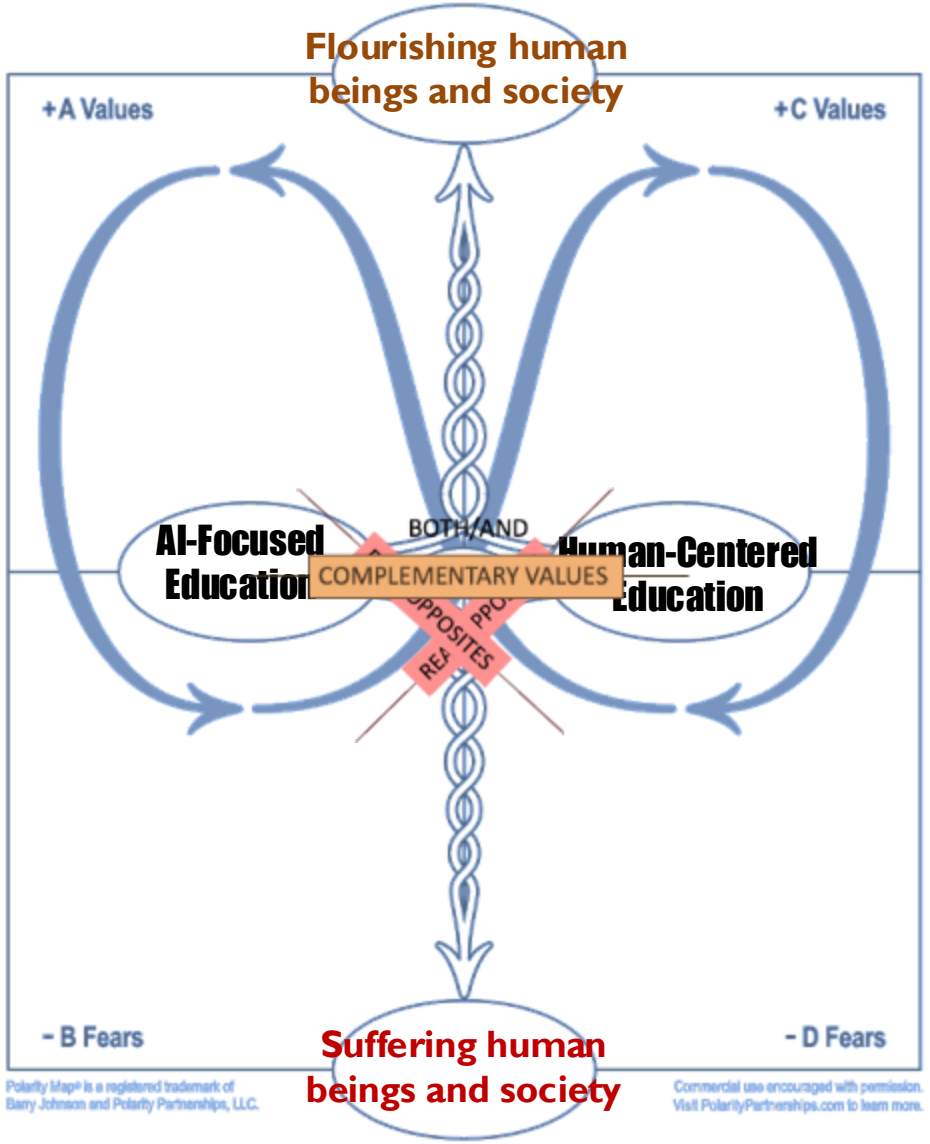


The Both-And Approach

What if we take a both-and approach?

- *Instant feedback 24/7*
- *Personalized learning at scale*
- *No bias or fatigue*
- *Consistent quality*
- *Global access*

- *Algorithmic bias*
- *Loss of human connection*
- *Cultural standardization*
- *Privacy concerns*
- *Hallucinations and errors*



*A **both-and approach** results in a WIN-WIN situation combining the best of both poles*

- *Inspiring human presence*
- *Deep mentoring relationships*
- *Cultural responsiveness*
- *Contextual wisdom*
- *Authentic assessment*

- *Overworked, exhausted faculty*
- *One-size-fits-all lectures*
- *Delayed, superficial feedback*
- *Unconscious bias*
- *Geographic inequity*

*An **either-or approach** results in a LOSE-LOSE situation combining the worst of both poles*

2

Overview of some projects



Implementing Custom GPT in a classroom

Qadir, J. *Generative AI in Undergraduate Classrooms: Lessons from Implementing a Customized GPT Chatbot for Learning Enhancement*. In 2025 IEEE Global Engineering Education Conference (EDUCON) (pp. 1-10). IEEE.

Complex Problem Solving using Agentic AI

Mushtaq, A., Naeem, R., Ghaznavi, I., Taj, I., Hashmi, I., & **Qadir, J**. *Harnessing Multi-Agent LLMs for Complex Engineering Problem-Solving: A Framework for Senior Design Projects*. In 2025 IEEE Global Engineering Education Conference (EDUCON).

ChatGPT's CustomGPT

November 6, 2023 Product

Introducing GPTs

You can now create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

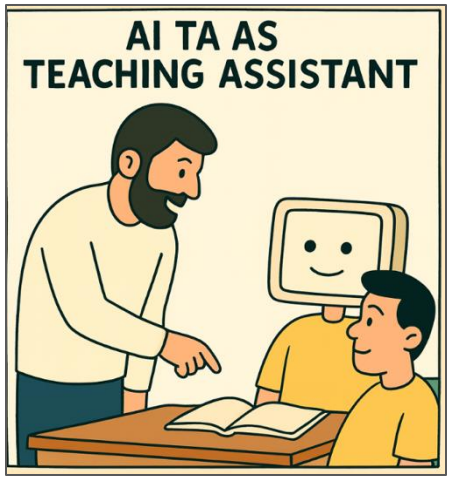
GPTs let you customize ChatGPT for a specific purpose

Since launching ChatGPT people have been asking for ways to customize ChatGPT to fit specific ways that they use it. We launched Custom Instructions in July that let you set some preferences, but requests for more control kept coming. Many power users maintain a list of carefully crafted prompts and instruction sets, manually copying them into ChatGPT. GPTs now do all of that for you.

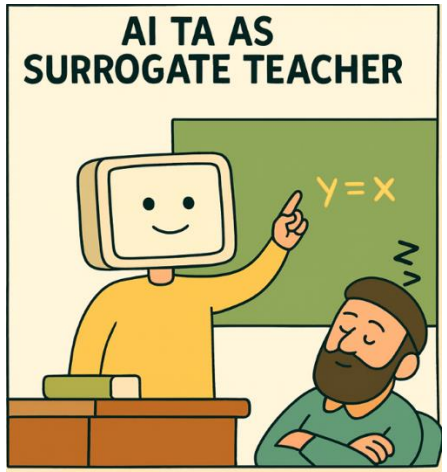


Various ways we can use AI in education

Role of an AI TA



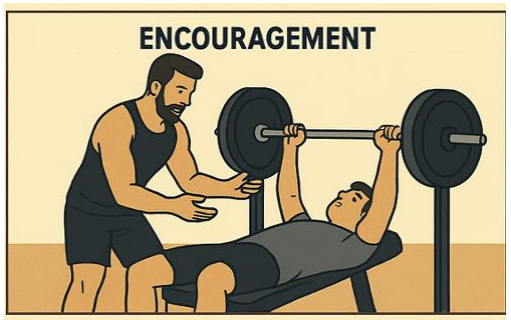
AUGMENTATION:
Supports the teacher by managing routine tasks, offering quick feedback, and extending classroom capacity.

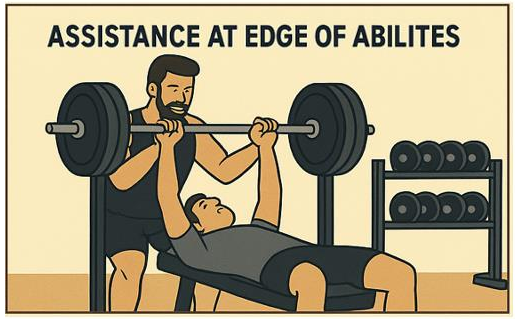


AUTOMATION:
Functions as a teaching machine—efficient, tireless, and unsettlingly capable of replacing the very human it was meant to assist.

The Role of a Coach









Variety of Customization Options

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:        ← task description
cheese =>                           ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
Translate English to French:        ← task description
sea otter => loutre de mer          ← example
cheese =>                           ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:        ← task description
sea otter => loutre de mer          ← examples
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>                           ← prompt
```

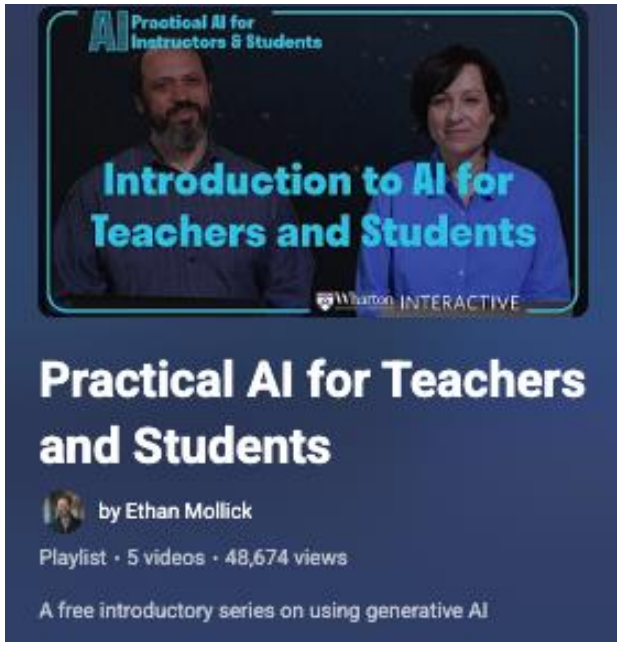
Lightweight (low cost, fast to adapt)
—**Zero-shot prompting** – works “out of the box.”
—**One-shot / Few-shot prompting** – A small number of examples guide the response style.

OTHER TECHNIQUES

Medium-weight (structured augmentation)
—**Retrieval-Augmented Generation (RAG)** – Injects external, up-to-date knowledge.
—**Agentic AI / Multi-Agent Systems** – Breaks tasks into specialized agents for robustness and depth.

Heavyweight (high cost, deeply specialized)
—**Fine-tuning** – Adjusts parameters on domain-specific data.
—**Training from scratch** – Builds a new model; most expensive but fully customizable.

Pedagogical tuning of ChatGPT



Role: Tell the AI who it is.

Example: *You are a friendly, helpful mentor who gives students advice and feedback about their work.*

Goal: Tell the AI what you want it to do.

Example: *Give students feedback on their [project outline, assignment] that takes the goal of the assignment into account and pinpoints specific ways they might improve the work.*

Step-by-step instructions.

Example: *Introduce yourself to the student as their mentor and ask them to share their work so that you can provide feedback. Wait for the student to respond. Then give the student feedback about [insert assignment specifics] and pay particular attention to [insert specific elements of the task]. Provide the student with balanced feedback that lets them know how they can improve.*

Add personalization.

Example: *This is a new project that students are working on. This is a first attempt at a proposed outline. General suggestions that address gaps, and missing steps, are helpful.*

Add your own constraints.

Example: Provide students with suggestions but not to actually revise the work.

CustomGPT in Different Roles

AI Role	Example Prompt and Additional Considerations
 **AI as Explainer**	***Prompt:*** "You generate clear, accurate examples for students of concepts. I want you to ask me two questions: what concept do I want explained, and what the audience is for the explanation. Provide a clear, multiple-paragraph explanation of the concept using specific examples and give me five analogies I can use to understand the concept in different ways." ***Customization Options:*** You can highlight your current level, language proficiency, or prior knowledge.
 **AI for Low-Stake Testing**	***Prompt:*** "You are a quiz creator of highly diagnostic quizzes. You will make good low-stakes tests and diagnostics. You will then ask me two questions: (1) what, specifically, should the quiz test; (2) who the audience is. Once I provide answers, you will construct multiple-choice questions that go beyond facts, including plausible alternatives without an "all of the above" option. At the end, you will provide an answer key with explanations." ***Customization Options:*** Specify content to be covered or difficulty level (easy, medium, advanced).
 **AI as Example Generator**	***Prompt:*** "I would like you to act as an example generator for students. When dealing with new and complex concepts, varied examples help students understand better. Ask me for the concept and student level, then provide four diverse, accurate examples of the concept in action." ***Customization Options:*** You can specify current level, language proficiency, prior knowledge, or preferred target domain for analogies.
 **AI for Formative Feedback**	***Prompt:*** "You are a constructive feedback assistant. When reviewing student responses or project drafts, start by asking me about the specific aspects I want feedback on (e.g., clarity, accuracy, depth). Then, provide thoughtful feedback addressing both strengths and areas for improvement. Offer actionable suggestions that guide students on how to enhance their work and deepen their understanding. Ensure feedback is specific, encouraging, and aligned with learning objectives." ***Customization Options:*** If a rubric is available, incorporate it to offer more targeted feedback.



جامعة قطر
QATAR UNIVERSITY
كلية الهندسة
COLLEGE OF ENGINEERING

Ethics Course
(77 Students: 46 users)

200+ Chats

CS200-Ethics-TA

By Junaid Qadir

A Socratic tutor for CS ethics, simplifying concepts in English-Arabic with course materials.

- Can you explain the ethical concerns of AI?
- What is your understanding of privacy issues in...
- What ethical dilemmas could arise with softwar...
- How should we approach the final project on...

Data Comm. & Networks-I Course
(62 Students, 61 users)

700+ Chats

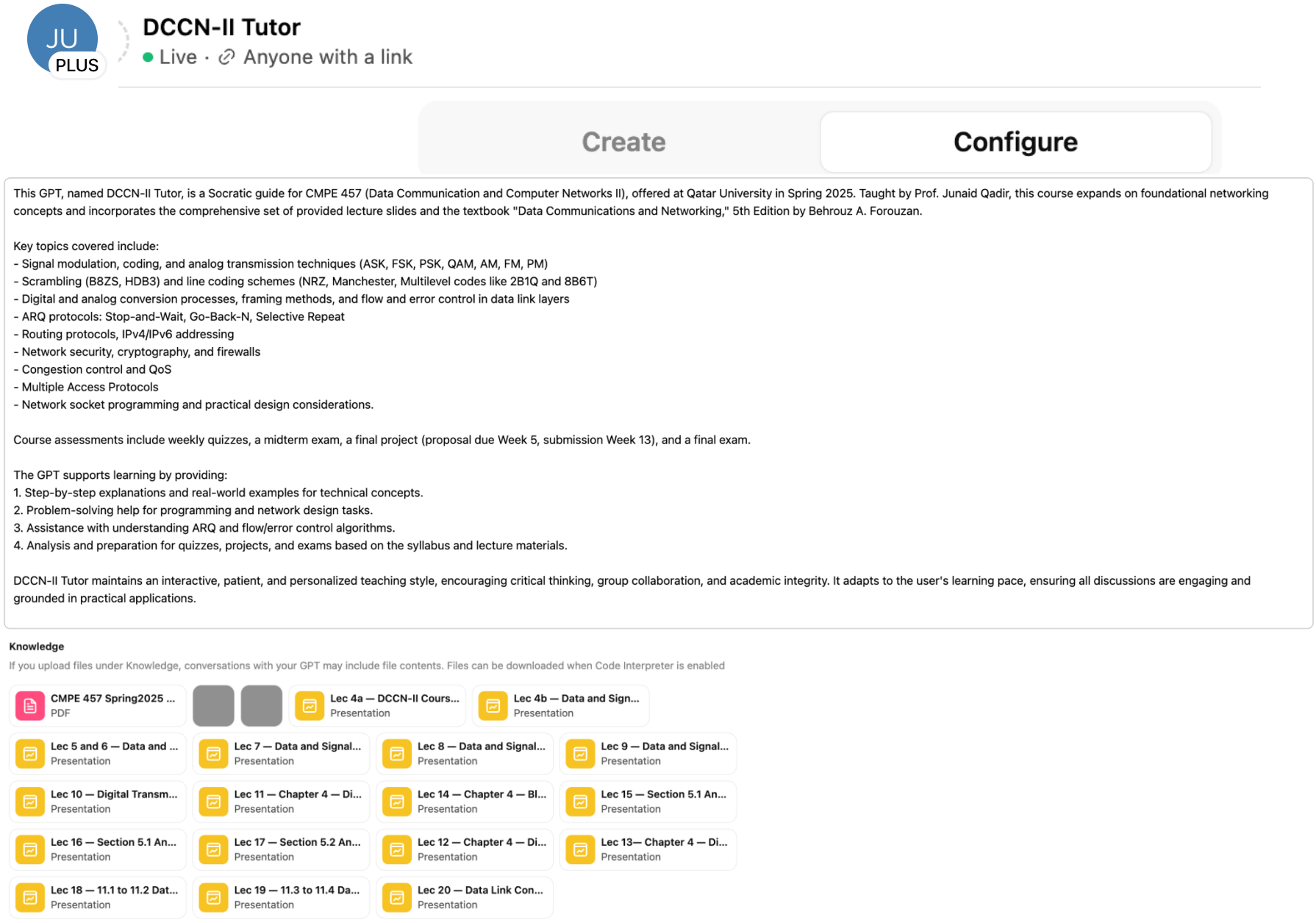
DCCN-II Tutor

By Junaid Qadir

A Socratic guide for CMPE 457 students at Qatar University.

- What are the advantages of using HDB3...
- Can you explain the Stop-and-Wait ARQ protocol wit...
- How does 4D-PAM5 improve bandwidt...
- What are the main differences between Go-Back...

The Customization of CustomGPT



Pre- and Post-Intervention Results for the Ethics Course (45 Students).

This table shows results for various learning outcomes before and after using the CustomGPT chatbot. Data are based on a 1-5 Likert scale, where 1 indicates a negative outcome, 3 is neutral, and 5 is positive.

Ethics Course (45 Students)

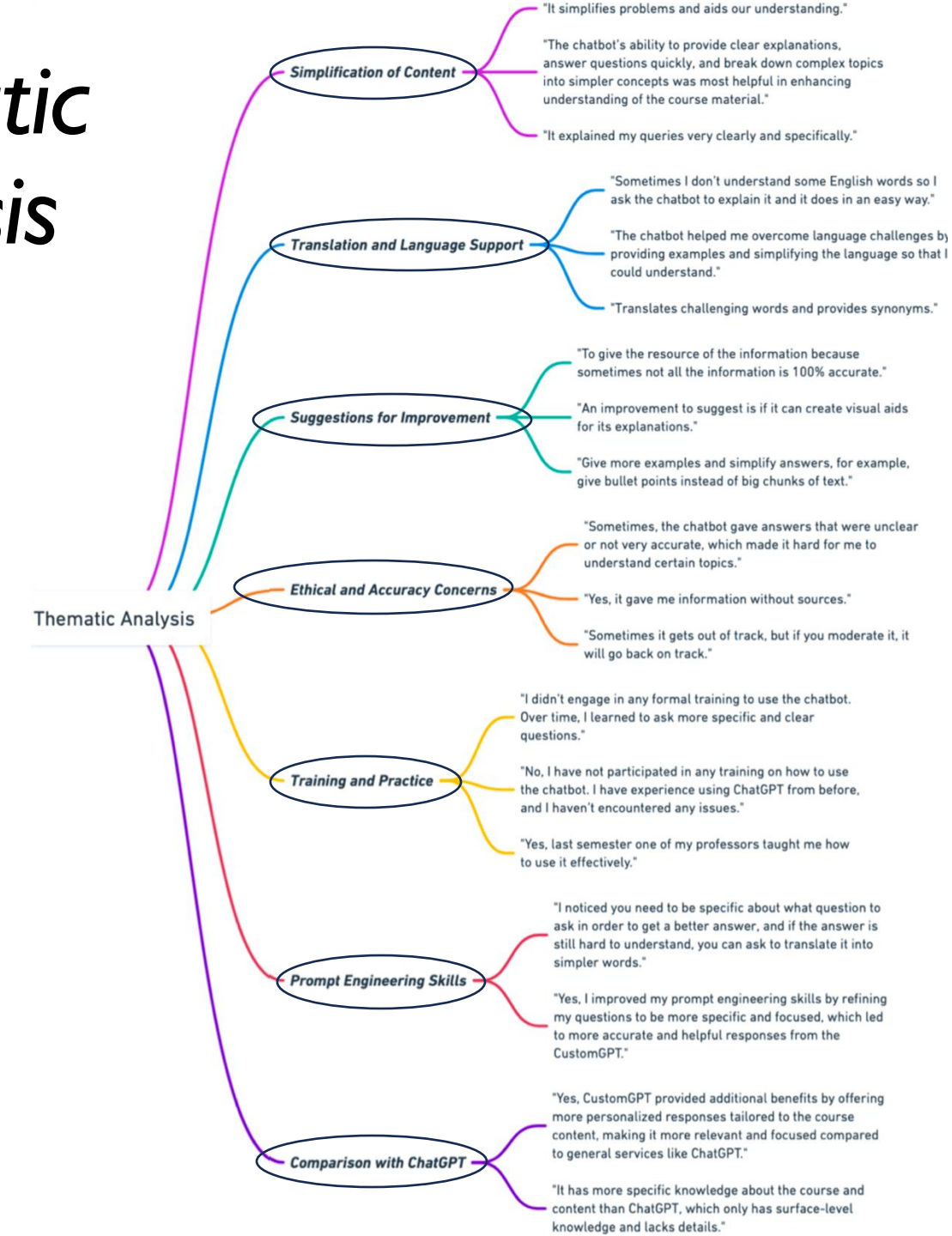
Metric	*Pre-Mean (SD)*	*Post-Mean (SD)*	*Mean Diff.*	*Pre-Mode*	*Post-Mode*	*t-statistic*	*p-value*	*Significant*
Confidence in Solving Problems	2.93 (0.86)	3.44 (1.02)	0.51	3	4	16.88	8.03e-15	Yes
Understanding Course Content	3.18 (0.81)	3.80 (0.90)	0.62	3	4	16.88	8.03e-15	Yes
Engagement with the Course	2.91 (0.91)	3.27 (1.07)	0.36	3	4	34.67	5.15e-22	Yes
Understanding Terminology	2.82 (0.89)	3.71 (1.07)	0.89	3	4	11.31	4.19e-11	Yes
Performance in Assessments	3.44 (0.89)	3.67 (1.00)	0.23	3	4	12.27	7.83e-12	Yes
Ease of Interaction	N/A	4.33 (N/A)	N/A	N/A	5	N/A	N/A	N/A

Pre- and Post-Intervention Results for the DCCN-I Course (61 Students).

DCCN-I Course (61 Students)

Metric	*Pre-Mean (SD)*	*Post-Mean (SD)*	*Mean Diff.*	*Pre-Mode*	*Post-Mode*	*t-statistic*	*p-value*	*Significant*
Confidence in Solving Problems	3.08 (1.01)	3.64 (0.99)	0.56	3	4	16.90	7.90e-15	Yes
Understanding Course Content	3.30 (0.97)	3.98 (0.97)	0.68	3	4	16.80	8.00e-15	Yes
Engagement with the Course	3.16 (0.99)	3.44 (1.08)	0.28	3	4	34.50	5.00e-22	Yes
Understanding Terminology	3.03 (1.00)	3.80 (1.16)	0.77	3	4	11.20	4.00e-11	Yes
Performance in Assessments	3.43 (0.93)	3.72 (0.93)	0.29	3	4	12.30	7.80e-12	Yes
Ease of Interaction	N/A	4.44 (N/A)	N/A	N/A	5	N/A	N/A	N/A

Thematic Analysis



Limitations

— The tool use was encouraged but not directly part of the classroom pedagogy; thus its use depended on the students (some students did not use it at all)

— Students could only be minimally trained on how to use the tool optimally.

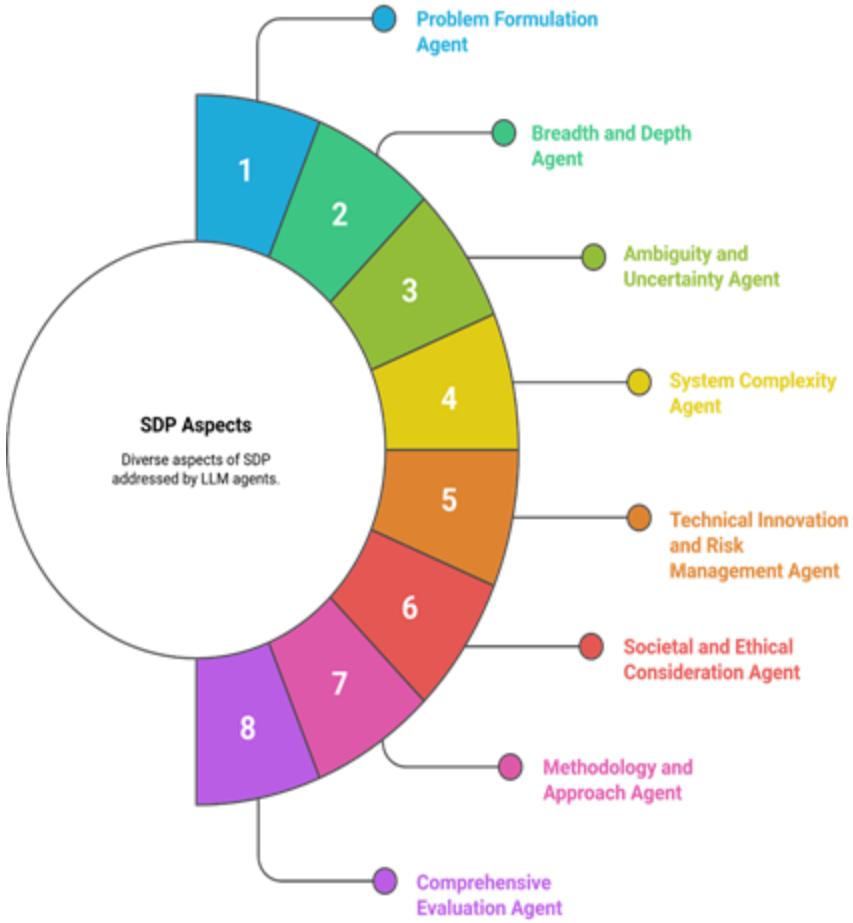
— The study relied solely on self-reported questionnaire data, without usage tracking or triangulation through interviews or focus groups.

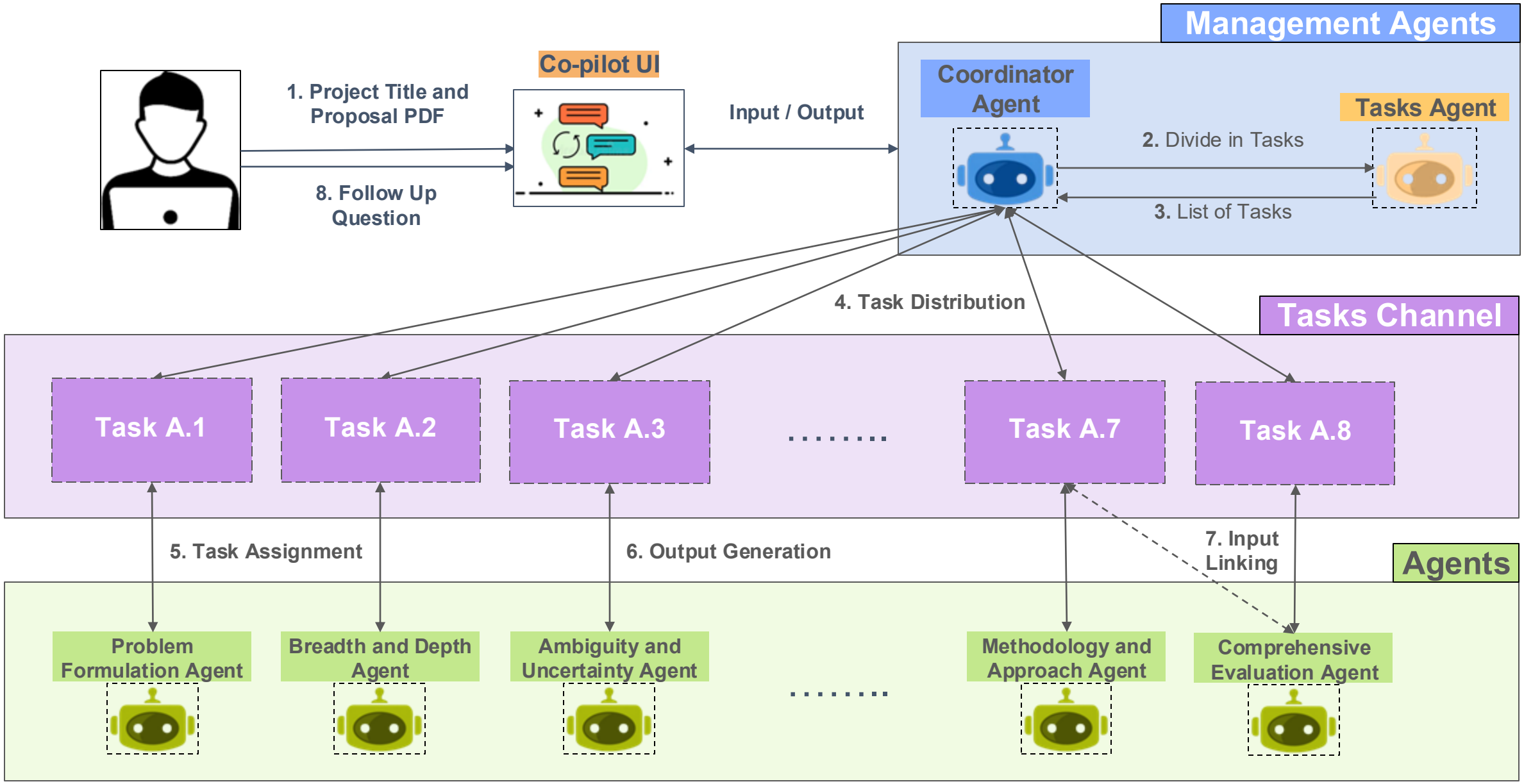
— This was a single-author study with no external validation; To refine question design and reduce bias, reflexivity and iterative AI-assisted feedback were used.

IEEE EDUCON 2025
IEEE Global Engineering Education Conference
London, United Kingdom || 22-25 April 2025 || Queen Mary University of London

Complex Problem Solving using Agentic AI

Mushtaq, A., Naeem, R., Ghaznavi, I., Taj, I., Hashmi, I., & **Qadir, J**. *Harnessing Multi-Agent LLMs for Complex Engineering Problem-Solving: A Framework for Senior Design Projects*. In 2025 IEEE Global Engineering Education Conference (EDUCON).




Management Agents
Co-pilot UI
Coordinator Agent
Tasks Agent
1. Project Title and Proposal PDF
8. Follow Up Question
Input / Output
2. Divide in Tasks
3. List of Tasks
4. Task Distribution
Tasks Channel
Task A.1
Task A.2
Task A.3
. . . . . .
Task A.7
Task A.8
5. Task Assignment
6. Output Generation
7. Input Linking
Agents
Problem Formulation Agent
Breadth and Depth Agent
Ambiguity and Uncertainty Agent
. . . . . .
Methodology and Approach Agent
Comprehensive Evaluation Agent

#	Agents	Details	
1	Problem Formulation Agent	Description	You are an AI assistant and expert in evaluation of Problem Formulation part of the Engineering Senior Year Design Projects.
		Task	Assess the clarity, depth, and scope of the problem statement
		Objective	Determine if the problem is appropriately complex and well-defined
		Evaluation Points	* Does the proposal clearly articulate the engineering problem? * Is the complexity level of the problem aligned with advanced engineering challenges? * Does it highlight any interdisciplinary aspects or requirements that add to the complexity? * Are the societal, environmental, or ethical implications of the problem considered in the formulation?
2	Breadth and Depth Agent	Description	You are an AI assistant and expert in evaluation of Breadth and Depth part of the Engineering Senior Year Design Projects.
		Task	Evaluate the breadth (interdisciplinary aspects) and depth (in-depth analysis and specialized knowledge) required for solving the problem.
		Objective	Check if the project goes beyond routine tasks and requires comprehensive engineering knowledge.
		Evaluation Points	* Does the proposal show evidence of requiring interdisciplinary knowledge (e.g., integration of mechanical, electrical, or software engineering principles)? * Are specialized techniques, theories, or methods needed for this project? * Is there evidence of in-depth analysis, such as detailed background research or literature review, that supports problem formulation?
3	Ambiguity and Uncertainty Agent	Description	You are an AI assistant and expert in evaluation of Ambiguity and Uncertainty part of the Engineering Senior Year Design Projects.
		Task	Identify areas in the proposal where uncertainty or ambiguity is present.
		Objective	Assess whether the project includes uncertain elements and requires assumptions or estimations.
		Evaluation Points	* Are there data gaps or ambiguous elements in the problem that require assumptions or approximations? * Does the proposal acknowledge potential sources of uncertainty or unknown variables? * How well does the project formulation plan to address and manage this ambiguity?
4	System Complexity Agent	Description	You are an AI assistant and expert in evaluation of System Complexity part of the Engineering Senior Year Design Projects.
		Task	Analyze the system complexity within the proposal.
		Objective	Evaluate whether the project involves multiple, interconnected components that add complexity.
		Evaluation Points	* Does the problem involve managing interactions between multiple subsystems? * Are there dependencies or integrations that need special attention? * Is there a structured approach for managing these complex system interactions, such as a modular design or layered architecture?
5	Technical Innovation and Risk Management Agent	Description	You are an AI assistant and expert in evaluation of Technical Innovation and Risk Management part of the Engineering Senior Year Design Projects.
		Task	Evaluate the project's innovation level and technical unpredictability.
		Objective	Determine if the project aims to push boundaries with novel approaches or technologies.
		Evaluation Points	* Does the proposal include innovative solutions or cutting-edge technology that is technically challenging? * Are existing solutions insufficient, requiring new methods or adaptations? * Does the project plan account for or manage technical unpredictability?
6	Societal and Ethical Consideration Agent	Description	You are an AI assistant and expert in evaluation of Societal and Ethical Consideration part of the Engineering Senior Year Design Projects.
		Task	Assess the attention given to societal, environmental, and ethical factors.
		Objective	Verify if broader implications are acknowledged and addressed in the project.
		Evaluation Points	* Are societal, environmental, or ethical impacts explicitly considered in the project proposal? * Does the proposal outline steps to mitigate any adverse effects or ethical concerns? * Is there a justification for how the project aligns with public safety, welfare, or environmental goals?
7	Methodology and Approach Agent	Description	You are an AI assistant and expert in evaluation of Methodology and Approach part of the Engineering Senior Year Design Projects.
		Task	Critique the methodology and approach proposed for solving the problem.
		Objective	Ensure that the methodology aligns with solving complex engineering problems and leverages appropriate analytical tools.
		Evaluation Points	* Is the proposed methodology rigorous and well-suited for handling the identified complexity? * Are advanced analytical tools, simulations, or modeling approaches specified? * Does the methodology account for iterative testing, prototyping, or validation against real-world conditions?
8	Comprehensive Evaluation Agent	Description	You are an AI assistant and expert in evaluation of Comprehensive Evaluation part of the Engineering Senior Year Design Projects.
		Task	Synthesize evaluations from all other agents to provide an overall score and summary.
		Objective	Offer a holistic assessment of the project's formulation, analysis, and methodology.
		Evaluation Points	* How well does the project proposal align with ABET's definition of complex engineering problems? * What are the overall strengths and weaknesses based on agent feedback? * Are there any key areas for improvement identified across multiple agents?

Multi-Agents Powered Senior Design Project Co-Pilot

Input Section

Enter Title

Upload Proposal PDF

Drop File Here
- or -
Click to Upload

Result

Title and PDF received. Initial feedback provided by agents.

Submit

Followup Message

Ask Follow-up Questions After Initial Feedback from

Send Message

Multi-Agent System Responses and Scores

Summarized Output

Project Title: ...

Overall Score: 3.5/5 (Average Score Across All Agents)

Evaluation Summary:
The project proposal is well-formulated and demonstrates a strong approach to tackling a complex engineering problem. However, there are areas for improvement in the depth of analysis, ...

Detailed Scores and Feedback by Criterion:

Overall Comments: The project is well-aligned with complex engineering problem-solving...

Agent-Wise Responses

Agent: Problem Formulation Agent
Evaluation of Problem Formulation

Agent: Breadth and Depth Agent
Evaluation of Breadth and Depth of Research

Agent: Ambiguity and Uncertainty Agent
Evaluation of Ambiguity and Uncertainty

Agent: System Complexity Agent
Evaluation of System Complexity

Agent: Technical Innovation and Risk Management Agent
Evaluation of Technical Innovation and Risk Management

Agent: Societal and Ethical Consideration Agent
Evaluation of Societal and Ethical Considerations

Agent: Methodology and Approach Agent
Evaluation of Methodology and Approach

Agent: Comprehensive Evaluation Agent
Comprehensive Evaluation of the Proposal

NLP-based Scores

Calculated using the Submitted Proposal

These are derived form the submitted proposals. Scores from Agents will be displayed in the middle section.

Lexical Cohesion

Score: 0.27

Measures thematic consistency by analyzing word repetition or related terms, indicating how well the content in the proposal is built on multiple ideas. Scores range from 0 (no cohesion) to 1 (full thematic consistency).

Clause Density

Score: 2.7

Captures sentence complexity by measuring clauses per sentence, reflecting layered perspectives. Scores range from 1 (simple, single-idea sentences) to 3+ (highly complex, multi-idea sentences)

Flesch-Kincaid

Score: 14

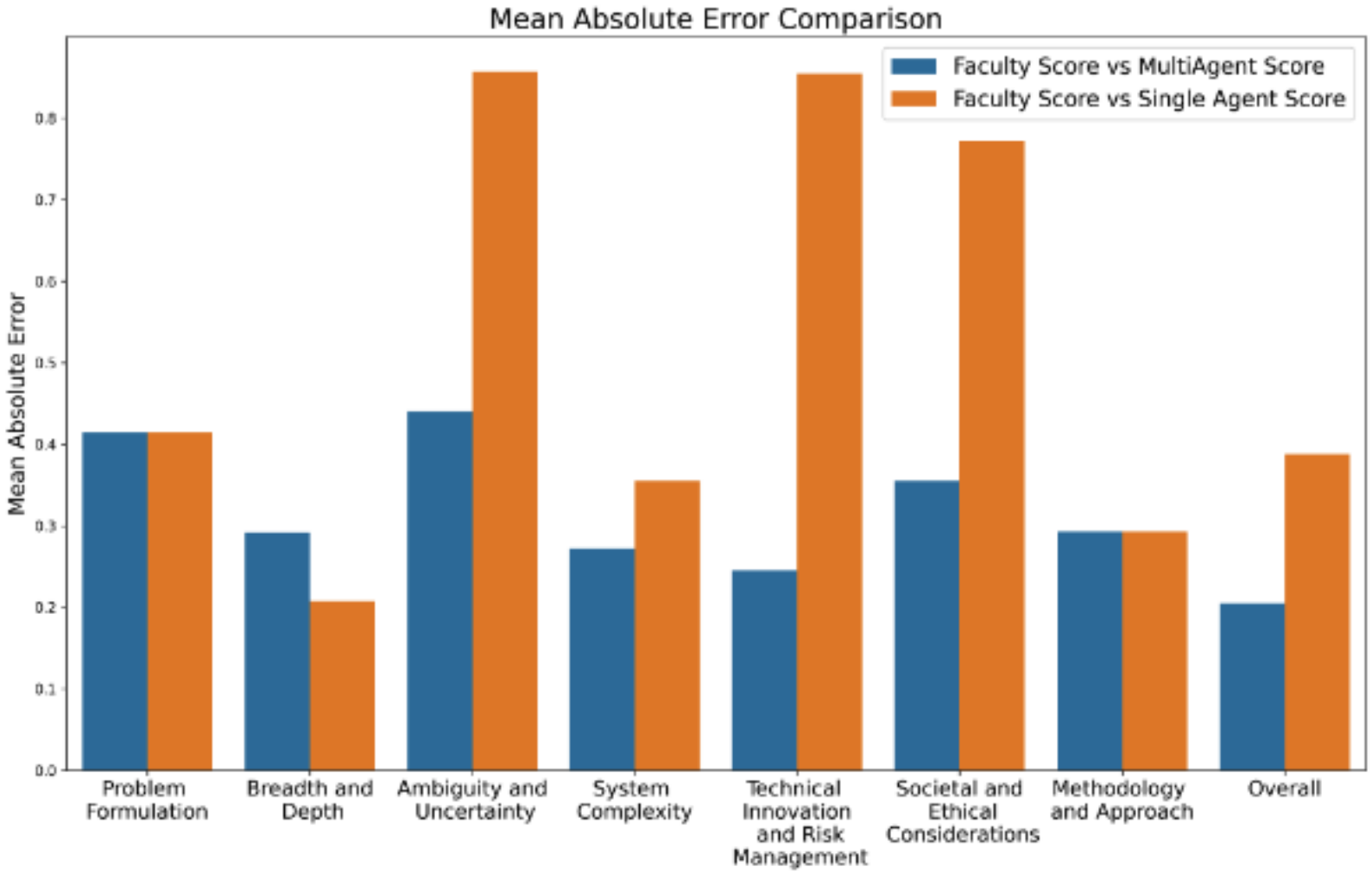
Estimates readability, indicating the U.S. grade level needed to understand the text for first time. An ideal range balancing accessibility and sophistication (0–16 scale) for bachelor's-level academic purposes.

Avg Sentence Length

Score: 16.8

Indicates structural complexity and content depth. Shorter sentences enhance readability, while longer ones may reflect richer perspectives but harder to follow.

Use via API · Built with Gradio

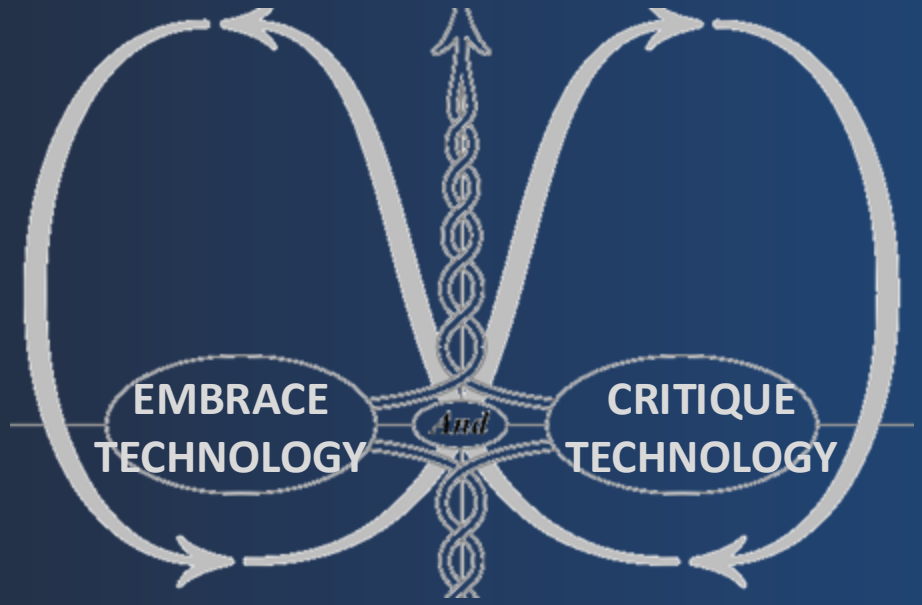


Comparison of faculty evaluations with multi-agent and single-agent systems.

The multi-agent system (MAS) provides feedback that is not only quantitatively more aligned with faculty scores but also qualitatively more comprehensive and detailed, often mirroring the depth of faculty evaluations. This shows that multi-agent LLMs can deliver richer, more reliable feedback than single-agent approaches.

3

Talking Points/ Summary



Both-And Wisdom-Based Education

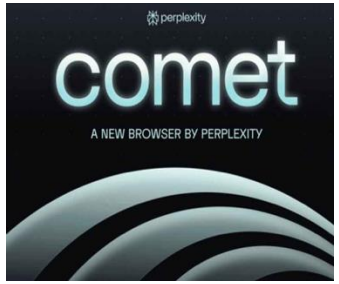
Promise and risks with Agentic AI



AGENTIC CODING

You can create your own educational game (without any programming skills).

multiple-choice quiz, drag & drop, simulation, branching scenario, gamified task, peer-review activity.





AGENTIC BROWSING

Can navigate across websites, extract data, compare results, and run mini-workflows

Can interact directly with online platforms — filling forms, submitting assignments, even completing quizzes.

Guidelines for Gen-AI Augmented Education



GROUNDED: To be able to use GenAI in Education, it is vital to **ensure as a designer and learner** that it is reliable, trustworthy, and grounded in facts

HUMANE: Education is a uniquely human process, requiring complex social interactions and extended period of enculturation.



INTENTIONAL: Efficient learning means refusing passivity, engaging, exploring, and actively generating hypotheses and testing them on the outside world.



DIALOGIC: Education is less about the content covered in classes and more about what an individual discovers and creates with available resources.



EXPERENTIAL: Don't automate what is experientially or pedagogically valuable to struggle through.



SELF-DEVELOPMENT: Education should aim to develop free, creative, and independent individuals who can appreciate culture & contribute to society.



Enable Relational and Collaborative AI Use
Incorporate pedagogy that support peer learning and dialogue to complement siloed use.



Design for Long-Term Learning Progression
Using AI to support the learning journey of the student continuously.

THE GUIDES FRAMEWORK:
Enhancing Engineering Education
With Generative AI

About this paper:

Appears in: EDULEARN24 Proceedings
Publication year: 2024
Pages: 8418-8428
ISBN: 978-84-09-62938-1
ISSN: 2340-1117
doi: 10.21125/edulearn.2024.2006

Looking forward to hearing your experience and insights

Happy to Collaborate 🤝

Thanks! jqadir@qu.edu.qa